

AIR UNIVERSITY, AEROSPACE AND AVIATION CAMPUS, KAMRA

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SUBMITTED TO	MAM MENAHIL NASEER
LAB	3
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Task # 1: Creating Different Types of NavBar

Build three small, separate Flutter screens (or one app with three routes) that each demonstrate a different navigation pattern:

• A Scaffold using only an AppBar with a title and two action icons (no Drawer, no BottomNavigationBar).

• A Scaffold using an AppBar + a Drawer with at least three ListTile destinations.

• A Scaffold using an AppBar + a BottomNavigationBar with three items, where tapping an item updates the body's content and the AppBar title.

Deliverable: a short screen recording or screenshots of all three nav patterns working, plus the source code.

CODE:

```
import 'package:flutter/material.dart';
void main() => runApp(const Task1App());
class Task1App extends StatelessWidget {
  const Task1App({super.key});
  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      debugShowCheckedModeBanner: false,
      home: const MenuScreen(),
    );
  }
}
class MenuScreen extends StatelessWidget {
  const MenuScreen({super.key});
  @override
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(title: const Text('Task 1: NavBar Patterns')),
      body: Center(
        child: Column(
```

```
        mainAxisAlignment: MainAxisAlignment.center,
        children: [
          ElevatedButton(
            onPressed: () => Navigator.push(
              context,
              MaterialPageRoute(builder: (_) => const AppBarOnlyScreen()),
            ),
            child: const Text('AppBar Only'),
          ),
          const SizedBox(height: 16),
          ElevatedButton(
            onPressed: () => Navigator.push(
              context,
              MaterialPageRoute(builder: (_) => const DrawerScreen()),
            ),
            child: const Text('AppBar + Drawer'),
          ),
          const SizedBox(height: 16),
          ElevatedButton(
            onPressed: () => Navigator.push(
              context,
              MaterialPageRoute(builder: (_) => const BottomNavScreen()),
            ),
            child: const Text('AppBar + BottomNavigationBar'),
          ),
        ],
      ),
    ),
  );
 }
}
```

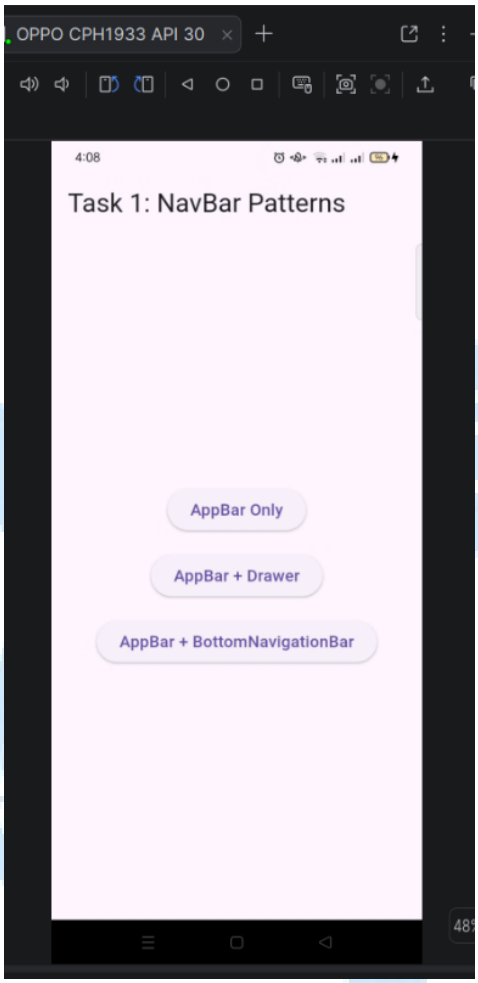
```
class AppBarOnlyScreen extends StatelessWidget {
 const AppBarOnlyScreen({super.key});
 @override
 Widget build(BuildContext context) {
  return Scaffold(
   appBar: AppBar(
    title: const Text('AppBar Only'),
    actions: [
     IconButton(icon: const Icon(Icons.search), onPressed: () {}),
     IconButton(icon: const Icon(Icons.more_vert), onPressed: () {}),
    ],
   ),
   body: const Center(child: Text('No Drawer, no BottomNavigationBar')),
  );
 }
}
class DrawerScreen extends StatelessWidget {
 const DrawerScreen({super.key});
 @override
 Widget build(BuildContext context) {
  return Scaffold(
   appBar: AppBar(title: const Text('AppBar + Drawer')),
   drawer: Drawer(
    child: ListView(
     padding: EdgeInsets.zero,
     children: [
      const DrawerHeader(
       decoration: BoxDecoration(color: Colors.indigo),
       child: Text(
        'Menu',
        style: TextStyle(color: Colors.white, fontSize: 24),
```

```
            ),
          ),
          ListTile(
            leading: const Icon(Icons.home),
            title: const Text('Home'),
            onTap: () => Navigator.pop(context),
          ),
          ListTile(
            leading: const Icon(Icons.info),
            title: const Text('About'),
            onTap: () => Navigator.pop(context),
          ),
          ListTile(
            leading: const Icon(Icons.settings),
            title: const Text('Settings'),
            onTap: () => Navigator.pop(context),
          ),
        ],
      ),
    ),
    body: const Center(child: Text('Open the Drawer from the top-left icon')),
  );
 }
}
class BottomNavScreen extends StatefulWidget {
 const BottomNavScreen({super.key});
 @override
 State<BottomNavScreen> createState() => _BottomNavScreenState();
}
class _BottomNavScreenState extends State<BottomNavScreen> {
 int _index = 0;
```

```
  final _titles = ['Home', 'Profile', 'Statistics'];
  final _bodies = const [
    Center(child: Text('Home Page')),
    Center(child: Text('Profile Page')),
    Center(child: Text('Statistics Page')),
  ];
  @override
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(title: Text(_titles[_index])),
      body: _bodies[_index],
      bottomNavigationBar: BottomNavigationBar(
        currentIndex: _index,
        onTap: (i) => setState(() => _index = i),
        items: const [
          BottomNavigationBarItem(icon: Icon(Icons.home), label: 'Home'),
          BottomNavigationBarItem(icon: Icon(Icons.person), label: 'Profile'),
          BottomNavigationBarItem(icon: Icon(Icons.bar_chart), label: 'Statistics'),
        ],
      ),
    );
  }
}
```

OUTPUT:

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Task # 2: Fitness Tracker App Main Structure

You are developing a multi-feature fitness tracker app using Flutter. Implement the main structure of the app using a Scaffold widget that includes an AppBar, Navigation Drawer, and Bottom Navigation Bar.

• AppBar: display the title of the current page and include action buttons for notifications and settings.

• Navigation Drawer: allow users to navigate between the main features "Workout Tracker," "Diet Monitor," "Progress," and "Settings."

• Bottom Navigation Bar: provide quick access to the primary features "Home," "Profile," and "Statistics."

• Selecting a Drawer item or a BottomNavigationBar item must update the AppBar title and the body content to match the selected section.

CODE:

```
import 'package:flutter/material.dart';
void main() => runApp(const FitnessTrackerApp());
class FitnessTrackerApp extends StatelessWidget {
  const FitnessTrackerApp({super.key});

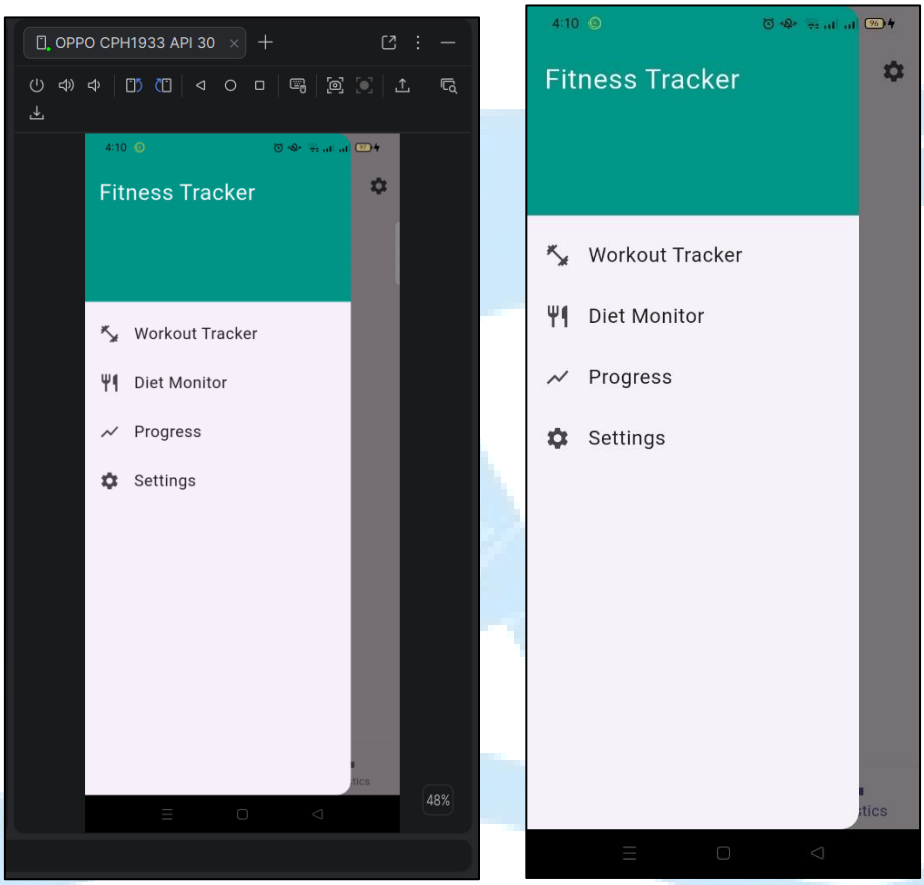

  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      debugShowCheckedModeBanner: false,
      home: const FitnessHome(),
    );
  }
}
class FitnessHome extends StatefulWidget {
  const FitnessHome({super.key});
  @override
  State<FitnessHome> createState() => _FitnessHomeState();
}
class _FitnessHomeState extends State<FitnessHome> {
  int _bottomIndex = 0;
  String _activeTitle = 'Home';
  final _bottomTitles = ['Home', 'Profile', 'Statistics'];
  final _drawerTitles = ['Workout Tracker', 'Diet Monitor', 'Progress', 'Settings'];
  void _selectFromDrawer(String title) {
    setState(() => _activeTitle = title);
    Navigator.pop(context);
  }
  void _selectFromBottomNav(int index) {
```

```
  setState(() {
    _bottomIndex = index;
    _activeTitle = _bottomTitles[index];
  });
}
@override
Widget build(BuildContext context) {
  return Scaffold(
    appBar: AppBar(
      title: Text(_activeTitle),
      actions: [
        IconButton(icon: const Icon(Icons.notifications), onPressed: () {}),
        IconButton(icon: const Icon(Icons.settings), onPressed: () {}),
      ],
    ),
    drawer: Drawer(
      child: ListView(
        padding: EdgeInsets.zero,
        children: [
          const DrawerHeader(
            decoration: BoxDecoration(color: Colors.teal),
            child: Text(
              'Fitness Tracker',
              style: TextStyle(color: Colors.white, fontSize: 22),
            ),
          ),
          ListTile(
            leading: const Icon(Icons.fitness_center),
            title: Text(_drawerTitles[0]),
            onTap: () => _selectFromDrawer(_drawerTitles[0]),
          ),
```

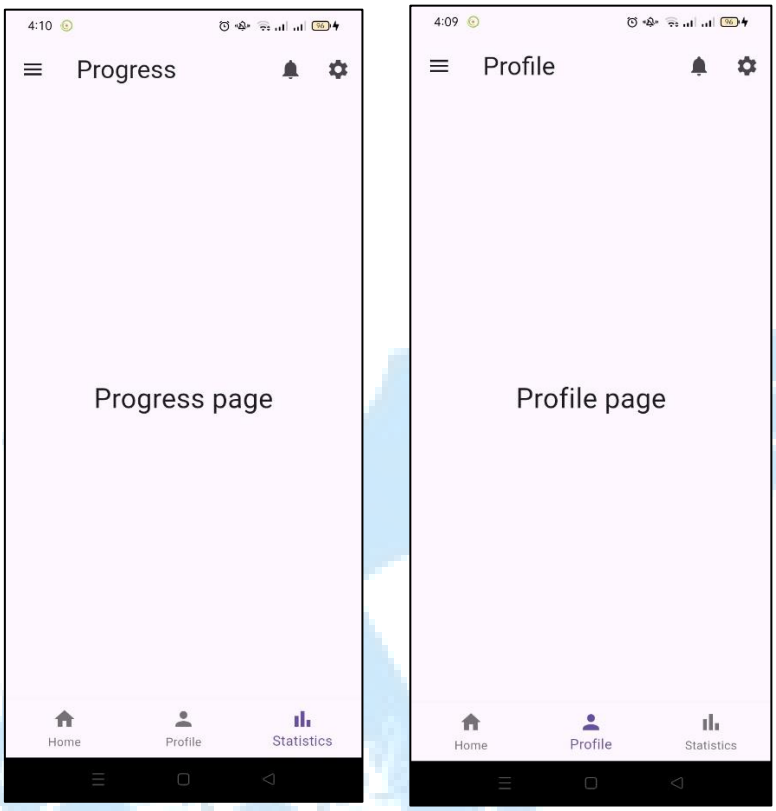
```
        ListTile(
          leading: const Icon(Icons.restaurant),
          title: Text(_drawerTitles[1]),
          onTap: () => _selectFromDrawer(_drawerTitles[1]),
        ),
        ListTile(
          leading: const Icon(Icons.show_chart),
          title: Text(_drawerTitles[2]),
          onTap: () => _selectFromDrawer(_drawerTitles[2]),
        ),
        ListTile(
          leading: const Icon(Icons.settings),
          title: Text(_drawerTitles[3]),
          onTap: () => _selectFromDrawer(_drawerTitles[3]),
        ),
      ],
    ),
  ),
  body: Center(
    child: Text('$_activeTitle page', style: const TextStyle(fontSize: 24)),
  ),
  bottomNavigationBar: BottomNavigationBar(
    currentIndex: _bottomIndex,
    onTap: _selectFromBottomNav,
    items: const [
      BottomNavigationBarItem(icon: Icon(Icons.home), label: 'Home'),
      BottomNavigationBarItem(icon: Icon(Icons.person), label: 'Profile'),
      BottomNavigationBarItem(icon: Icon(Icons.bar_chart), label: 'Statistics'),
    ],
  ),
);
```

```
 }
}
```

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Task # 3: Smart Home Control App Main Structure

You are tasked with developing a Smart Home Control app using Flutter. The app should let users monitor and control various aspects of their smart home. Create the main structure using a Scaffold widget that includes an AppBar, Navigation Drawer, and Bottom Navigation Bar.

• AppBar: display the title of the current control panel (e.g., "Lighting Control") and include action buttons for global notifications and quick settings.

• Navigation Drawer: allow users to navigate between control panels "Lighting," "Temperature," "Security," and

"Appliances."

• Bottom Navigation Bar: offer quick access to primary sections, "Dashboard," "Rooms," and "Settings."

• Each control panel selected from the Drawer should update the AppBar title to reflect the active panel.

CODE:

```
import 'package:flutter/material.dart';

void main() => runApp(const SmartHomeApp());

class SmartHomeApp extends StatelessWidget {
  const SmartHomeApp({super.key});

  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      debugShowCheckedModeBanner: false,
      home: const SmartHomeShell(),
    );
  }
}

class SmartHomeShell extends StatefulWidget {
  const SmartHomeShell({super.key});

  @override
  State<SmartHomeShell> createState() => _SmartHomeShellState();
}

class _SmartHomeShellState extends State<SmartHomeShell> {
  int _bottomIndex = 0;
  String _panelTitle = 'Lighting Control';

  final _bottomTitles = ['Dashboard', 'Rooms', 'Settings'];
  final _panels = const {
    'Lighting': 'Lighting Control',
```

```
  'Temperature': 'Temperature Control',
  'Security': 'Security Control',
  'Appliances': 'Appliances Control',
};
final _panelIcons = const {
  'Lighting': Icons.lightbulb,
  'Temperature': Icons.thermostat,
  'Security': Icons.security,
  'Appliances': Icons.kitchen,
};

void _selectPanel(String key) {
  setState(() => _panelTitle = _panels[key]!);
  Navigator.pop(context);
}

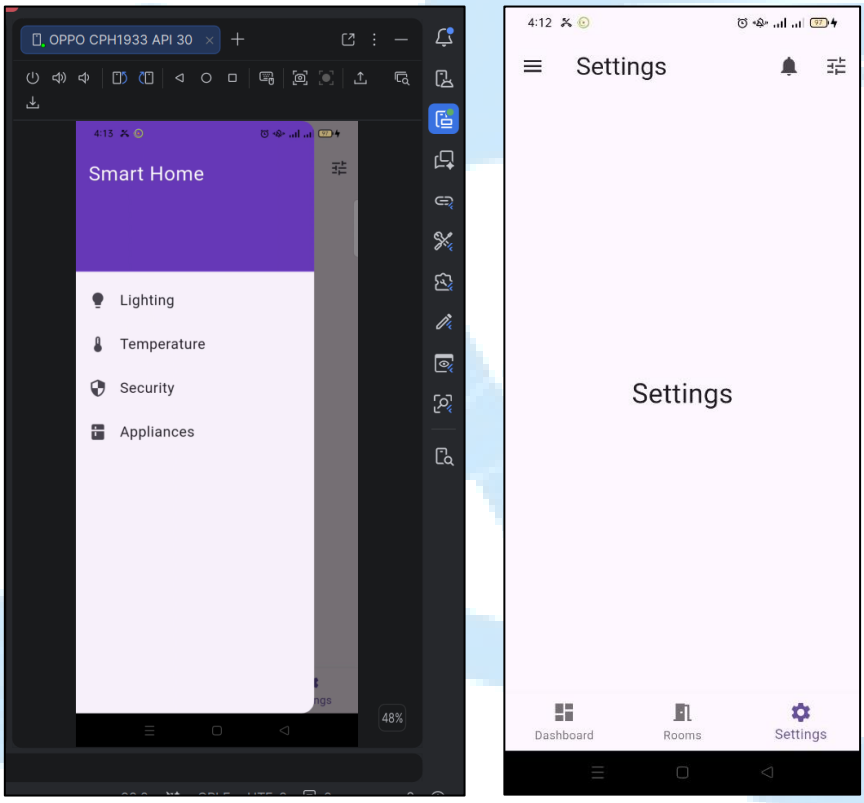
void _selectFromBottomNav(int index) {
  setState(() {
    _bottomIndex = index;
    _panelTitle = _bottomTitles[index];
  });
}

@override
Widget build(BuildContext context) {
  return Scaffold(
    appBar: AppBar(
      title: Text(_panelTitle),
      actions: [
        IconButton(icon: const Icon(Icons.notifications), onPressed: () {}),
        IconButton(icon: const Icon(Icons.tune), onPressed: () {}),
```

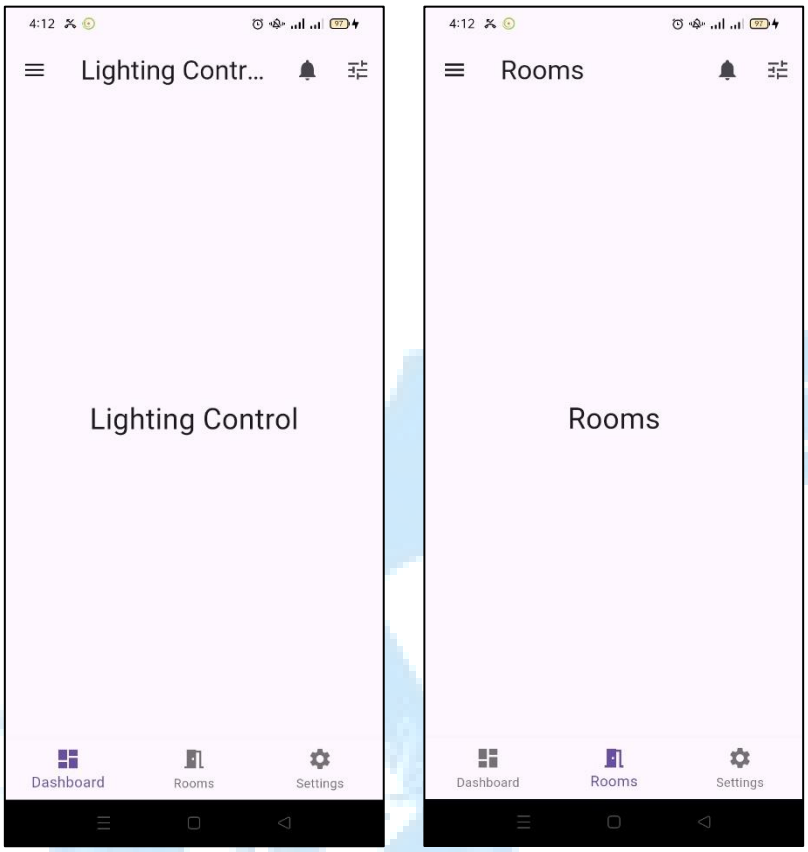
```
  ],
),
drawer: Drawer(
  child: ListView(
    padding: EdgeInsets.zero,
    children: [
      const DrawerHeader(
        decoration: BoxDecoration(color: Colors.deepPurple),
        child: Text(
          'Smart Home',
          style: TextStyle(color: Colors.white, fontSize: 22),
        ),
      ),
      for (final key in _panels.keys)
        ListTile(
          leading: Icon(_panelIcons[key]),
          title: Text(key),
          onTap: () => _selectPanel(key),
        ),
    ],
  ),
),
body: Center(
  child: Text(_panelTitle, style: const TextStyle(fontSize: 24)),
),
bottomNavigationBar: BottomNavigationBar(
  currentIndex: _bottomIndex,
  onTap: _selectFromBottomNav,
  items: const [
    BottomNavigationBarItem(icon: Icon(Icons.dashboard), label: 'Dashboard'),
    BottomNavigationBarItem(icon: Icon(Icons.meeting_room), label: 'Rooms'),
```

```
        BottomNavigationBarItem(icon: Icon(Icons.settings), label: 'Settings'),
      ],
    ),
  );
 }
}
```

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Task # 4: Design Your Own App (Open-Ended)

Pick any app idea of your own a recipe organizer, a music player, a study planner, a game hub, or anything else you like and build its main navigation structure using the same widgets covered in this lab (Scaffold, AppBar, Drawer, BottomNavigationBar).

You decide the details:

• What the app is, and what its sections/features are called.

• Which destinations belong in the Drawer versus the BottomNavigationBar, and why.

• At least one small enhancement beyond the minimum shown in class for example: icons that change color when selected, a Drawer header showing a “user profile” section, a notification badge/counter on an AppBar icon, or a FloatingActionButton that fits your app's theme.

Deliverable: working source code, plus a few sentences explaining which items you placed where and why. There is no single correct layout for this task, you are assessed on sound reasoning and correct use of the widgets, not on matching a template.

Design Explanation

App concept: a Recipe Organizer that lets a user browse recipes by meal category and save favorites.

Drawer vs BottomNavigationBar placement:

- The Drawer holds the four recipe categories (Breakfast, Lunch, Dinner, Favorites). These are content filters a user might switch between occasionally, not every screen tap, so they fit the "secondary, less-frequent navigation" role a Drawer is meant for.

- The BottomNavigationBar holds the three primary app-level destinations (Home, Search, Saved), since these are the handful of core actions a user returns to constantly and expects to reach in one tap.

Enhancements beyond the minimum:

1. Selected-state color: the active category's icon and label in the Drawer turn orange and bold, so the user can see

at a glance which filter is currently applied.

2. Notification badge: the AppBar's bookmark icon shows a small red counter badge with the number of saved recipes,

giving the user quick feedback without opening the Saved tab.

3. A FloatingActionButton (add icon) was added for quickly creating a new recipe, fitting the app's theme.

CODE:

```
import 'package:flutter/material.dart';
void main() => runApp(const RecipeOrganizerApp());
class RecipeOrganizerApp extends StatelessWidget {
 const RecipeOrganizerApp({super.key});
 @override
 Widget build(BuildContext context) {
  return MaterialApp(
    debugShowCheckedModeBanner: false,
    home: const RecipeHome(),
  );
 }
```

```
}
class RecipeHome extends StatefulWidget {
  const RecipeHome({super.key});
  @override
  State<RecipeHome> createState() => _RecipeHomeState();
}
class _RecipeHomeState extends State<RecipeHome> {
  static const Color lavender = Color(0xFFB57EDC);
  int _bottomIndex = 0;
  String _activeCategory = 'Home';
  final int _savedCount = 3;
  final _bottomTitles = ['Home', 'Search', 'Saved'];
  final _bottomIcons = [Icons.home, Icons.search, Icons.bookmark];
  final _categories = ['Breakfast', 'Lunch', 'Dinner', 'Favorites'];
  final _categoryIcons = [
    Icons.free_breakfast,
    Icons.lunch_dining,
    Icons.dinner_dining,
    Icons.favorite,
  ];
  void _selectCategory(String category) {
    setState(() => _activeCategory = category);
    Navigator.pop(context);
  }
  void _selectFromBottomNav(int index) {
    setState(() {
      _bottomIndex = index;
      _activeCategory = _bottomTitles[index];
    });
```

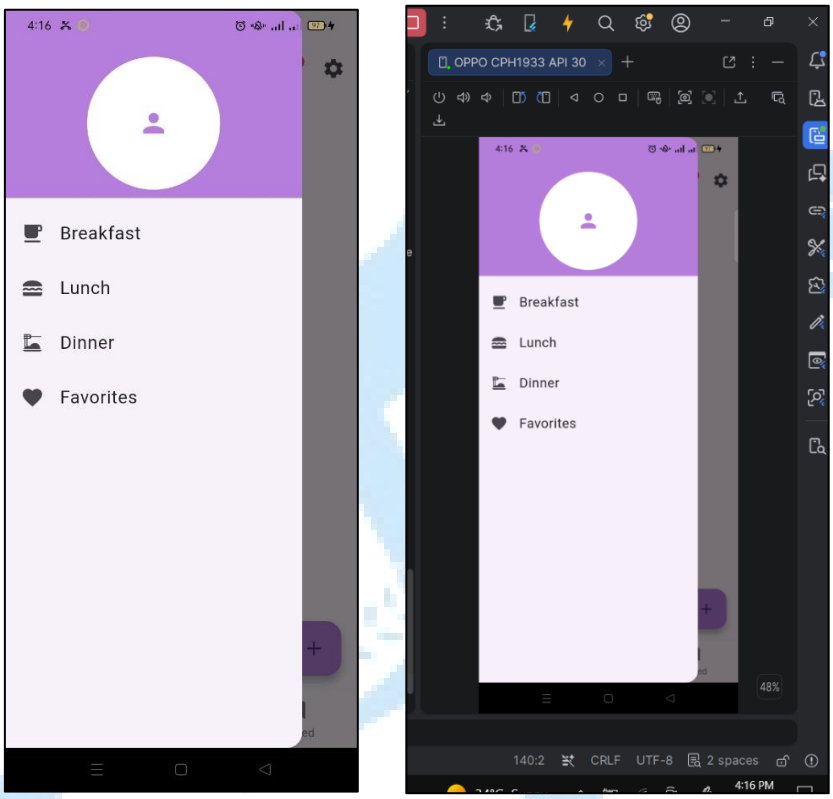
```
}
@override
Widget build(BuildContext context) {
  return Scaffold(
    appBar: AppBar(
      title: Text(_activeCategory),
      actions: [
        Stack(
          alignment: Alignment.center,
          children: [
            IconButton(icon: const Icon(Icons.bookmark), onPressed: () {}),
            if (_savedCount > 0)
              Positioned(
                right: 6,
                top: 6,
                child: Container(
                  padding: const EdgeInsets.all(3),
                  decoration: const BoxDecoration(
                    color: Colors.red,
                    shape: BoxShape.circle,
                  ),
                  constraints: const BoxConstraints(minWidth: 16, minHeight: 16),
                  child: Text(
                    '$_savedCount',
                    style: const TextStyle(color: Colors.white, fontSize: 10),
                    textAlign: TextAlign.center,
                  ),
                ),
              ),
```

```
        ],
      ),
      IconButton(icon: const Icon(Icons.settings), onPressed: () {}),
    ],
  ),
  drawer: Drawer(
    child: ListView(
      padding: EdgeInsets.zero,
      children: [
        const DrawerHeader(
          decoration: BoxDecoration(color: lavender),
          child: CircleAvatar(
            radius: 28,
            backgroundColor: Colors.white,
            child: Icon(Icons.person, size: 32, color: lavender),
          ),
        ),
        for (int i = 0; i < _categories.length; i++)
          ListTile(
            leading: Icon(
              _categoryIcons[i],
              color: _activeCategory == _categories[i] ? lavender : null,
            ),
            title: Text(
              _categories[i],
              style: TextStyle(
                color: _activeCategory == _categories[i] ? lavender : null,
                fontWeight: _activeCategory == _categories[i]
                    ? FontWeight.bold
```

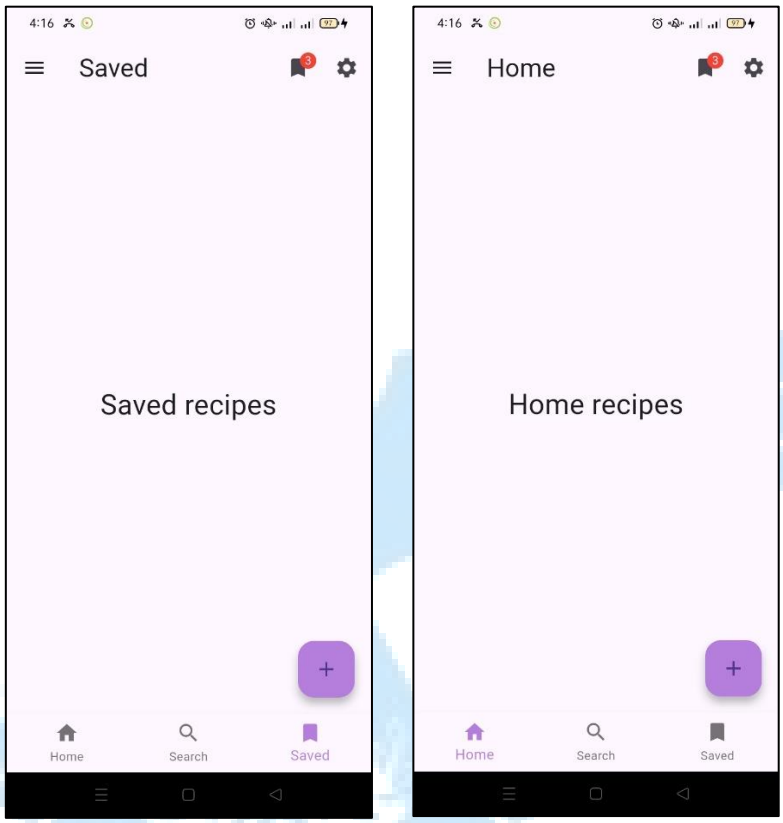
```
                    : FontWeight.normal,
              ),
            ),
            onTap: () => _selectCategory(_categories[i]),
          ),
        ],
      ),
    ),
    body: Center(
      child: Text('$_activeCategory recipes', style: const TextStyle(fontSize: 24)),
    ),
    floatingActionButton: FloatingActionButton(
      onPressed: () {},
      backgroundColor: lavender,
      child: const Icon(Icons.add),
    ),
    bottomNavigationBar: BottomNavigationBar(
      currentIndex: _bottomIndex,
      selectedItemColor: lavender,
      onTap: _selectFromBottomNav,
      items: [
        for (int i = 0; i < _bottomTitles.length; i++)
          BottomNavigationBarItem(
            icon: Icon(_bottomIcons[i]),
            label: _bottomTitles[i],
          ),
      ],
    ),
  );
```

```
 }
}
```

OUTPUT:



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